HR Analytics – Employee Attrition Report

Introduction

Employee attrition is a major concern for organizations as it affects productivity, increases recruitment costs, and disrupts workflow. Understanding the reasons behind employee turnover is essential for improving retention strategies. This project focuses on analyzing HR data to identify patterns and key factors influencing employee attrition using data analytics and machine learning techniques.

Abstract

This project aims to analyze employee data to understand the causes of attrition and predict future employee turnover. Exploratory Data Analysis (EDA) was performed to uncover trends related to salary, department, job role, and experience. Machine learning models such as Logistic Regression and Decision Tree were implemented to classify employees likely to leave. Additionally, SHAP (SHapley Additive Explanations) was used to interpret model predictions. Tableau was used to create interactive dashboards for visual analysis. The study reveals that factors such as low salary, lack of promotions, and job role significantly impact attrition.

Tools Used

- Python (Pandas, NumPy) – Data processing and manipulation
- Seaborn & Matplotlib – Data visualization
- Scikit-learn – Machine learning models
- SHAP – Model interpretability
- Tableau – Dashboard and data visualization
- VS Code – Development environment

Steps Involved in Building the Project

1. Collected and loaded the HR dataset
2. Performed data cleaning and preprocessing
3. Converted categorical variables using encoding techniques
4. Conducted Exploratory Data Analysis (EDA)
5. Identified important features affecting attrition
6. Split dataset into training and testing sets
7. Built classification models (Logistic Regression, Decision Tree)
8. Evaluated model performance using accuracy and classification report
9. Applied SHAP analysis to explain model predictions
10. Exported cleaned dataset and built interactive dashboards in Tableau

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Conclusion

The project successfully identified key factors influencing employee attrition, including salary, job role, and years at the company. Machine learning models provided accurate predictions, helping in identifying employees at risk of leaving. SHAP analysis improved the interpretability of the models by highlighting feature importance. The Tableau dashboard visually represented the insights, making it easier for decision-makers to understand trends and take necessary actions. Overall, this project demonstrates how data-driven approaches can help organizations reduce attrition and improve employee retention.

Summary

This project highlights the importance of combining data analytics, machine learning, and visualization tools to solve real-world HR problems. By identifying key attrition drivers, organizations can implement effective strategies to enhance employee satisfaction and retention.